

Predicting Life Expectancy Report

CSE437 Final Project Report

CSE437 — Group 1

September 2026

1. Introduction

1.1 Problem Statement

Life expectancy is one of the most widely used single-number summaries of a population's health and wellbeing, yet it is shaped by a wide mix of factors — healthcare access, immunization coverage, economic development, education, and disease burden among them. This project builds a regression model that predicts a country's life expectancy in a given year from a set of health, economic, and social indicators, and analyzes which of those indicators matter most and where the model's predictions can and cannot be trusted.

Target variable: life expectancy (years), a continuous variable — this is a **regression** problem.

1.2 Dataset

We use the **WHO Life Expectancy dataset**, originally compiled from the World Health Organization's Global Health Observatory and United Nations data, commonly distributed via Kaggle. It contains **2,938 rows and 22 columns**, covering **193 countries from 2000 to 2015** (one row per country-year). Columns include immunization rates (Hepatitis B, Polio, Diphtheria), mortality statistics (adult mortality, infant deaths, under-five deaths), economic indicators (GDP, percentage expenditure on health, income composition of resources), and social indicators (schooling, alcohol consumption, BMI).

This dataset was chosen because it is **real and messy**: missing values are common and unevenly distributed, several columns are heavily skewed, and column names/spacing are inconsistent in the raw file — it required genuine cleaning decisions rather than being ready to model out of the box.

1.3 Three Questions We Want the Data to Answer

1. **How much does healthcare expenditure (as % of GDP) actually correlate with life expectancy vs. schooling years?**
 2. **Do Developing vs. Developed status countries show different key predictors of life expectancy?**
 3. **Which single factor — immunization rates, BMI, or HIV/AIDS prevalence — has the largest negative effect on life expectancy?**
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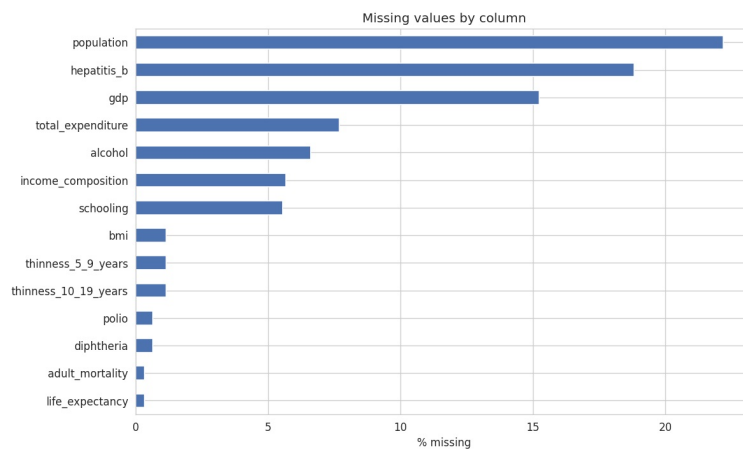
2. Data Handling and Preprocessing

Full detail and code: notebooks/01_data_audit_and_edu.ipynb,
notebooks/02_preprocessing.ipynb.

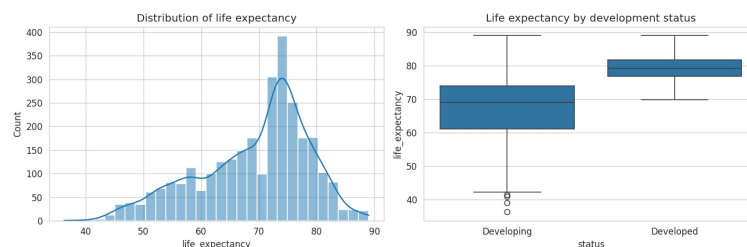
2.1 Data Audit

Before changing anything, we audited the raw data for structure, duplicates, and missingness.

- No exact duplicate rows; no duplicate (country, year) pairs.
- **Missingness is substantial and structural, not random:** population (22.2%), hepatitis_b (18.8%), gdp (15.2%), total_expenditure (7.7%), and several others have meaningful missing rates, and the missing rows cluster by country — some countries simply have gaps in specific indicators across most or all of their years on record (Figure 1).
- **19 rows have life expectancy below 45** — initially flagged as possible outliers, but verified against real-world events: Haiti (2010, the earthquake year), and Malawi, Sierra Leone, Zambia, and Zimbabwe in the early 2000s (the peak of the southern-African HIV/AIDS epidemic). These are kept as genuine signal, not removed.
- Several count/monetary features (measles, infant_deaths, under_five_deaths, population, gdp, percentage_expenditure) are heavily right-skewed.



Missing values by column



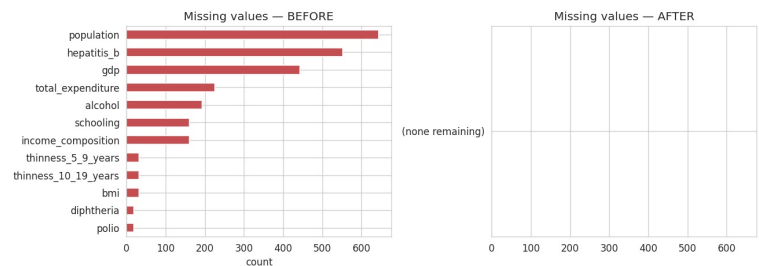
Distribution of the target variable, and by development status

2.2 Missing Value Imputation

A single global mean/median would blur real cross-country differences, since missingness clusters by country. We used a three-pass strategy:

1. **Within-country linear interpolation across years** — fills gaps between two known values for the same country.
2. **Within-country forward/backward fill** — covers missing values at the first or last year on record, which interpolation can't reach.
3. **Group median by development status** (Developed / Developing) — for the rare case where a country is missing a feature in *every* year, falling back to its peer group's median.

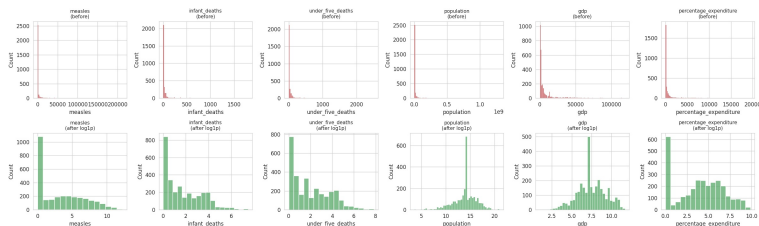
This reduced missing values from a maximum of 34.3% in a single column to **zero missing values** in the final dataset (Figure 2).



Missing values before vs. after imputation

2.3 Outlier and Skew Treatment

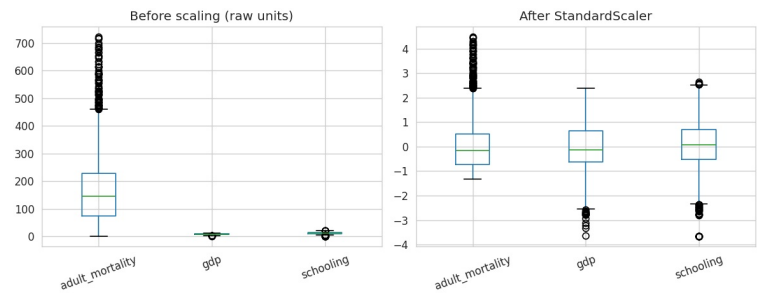
The low-life-expectancy rows discussed above were **kept** since they reflect real historical events. The six heavily right-skewed count/monetary columns were transformed with `log1p` (rather than deleted or capped), which visibly normalizes their distributions (Figure 3) without discarding any data.



Skewed features before and after log1p transform

2.4 Encoding and Scaling

The single categorical column, `status` (Developed/Developing), was binary-encoded to `status_developed`. Scaling (`StandardScaler`) was demonstrated on a sample of features to show its effect (Figure 4), but deliberately **not** applied to the whole dataset up front — instead it is fit inside each cross-validation fold during modeling, to avoid leaking test-set statistics into training.

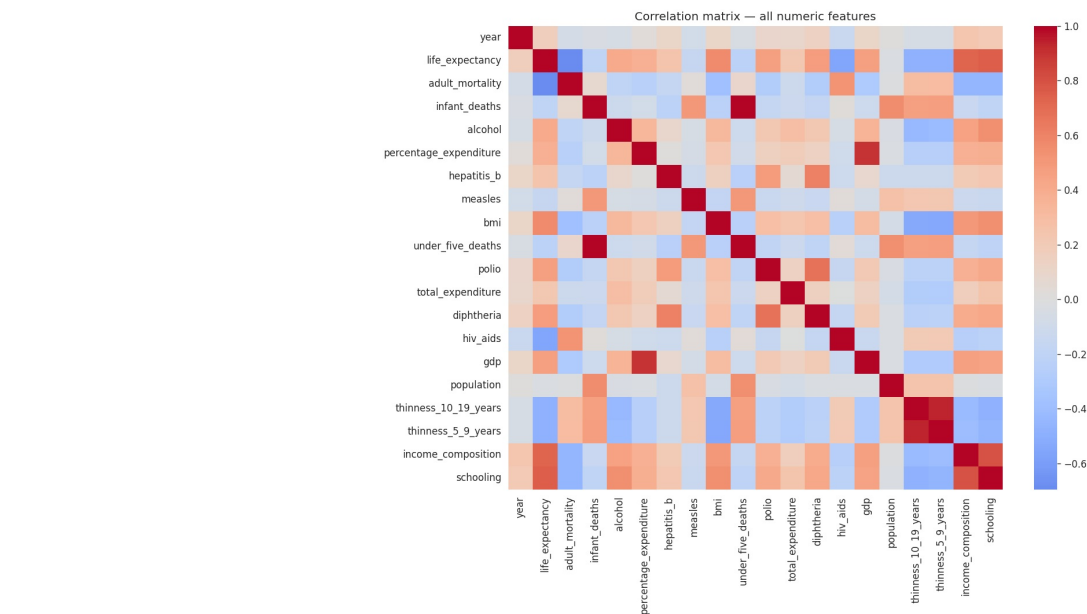


Effect of StandardScaler on three example features

3. Statistical Analysis

Descriptive statistics for every numeric column were computed in notebook 01 (`df.describe()`), covering mean, standard deviation, quartiles, and range. Key findings:

- Life expectancy ranges from 36.3 to 89 years (mean ≈ 69.2), left-skewed with a long low tail.
- `adult_mortality`, `income_composition`, and `schooling` show the strongest linear correlation with the target among raw features (Figure 5); `hiv_aids` and `adult_mortality` are also correlated with each other, an early signal of multicollinearity investigated further in Section 4.



Correlation matrix, all numeric features

4. Feature Engineering

Full detail: [notebooks/03_feature_engineering.ipynb](#).

4.1 Removing Redundant Features

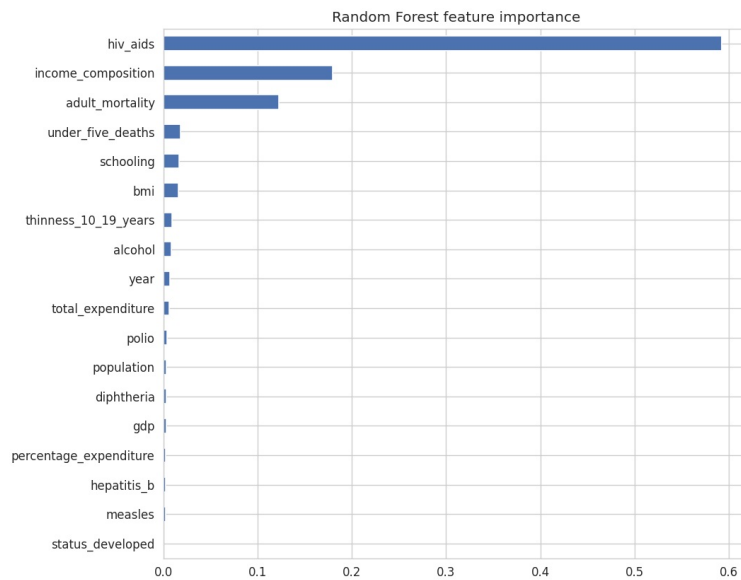
Two feature pairs were found to be near-duplicates (Pearson $r > 0.85$):

- `infant_deaths` and `under_five_deaths` ($r = 0.996$) — the latter structurally includes the former.
- `thinness_10_19_years` and `thinness_5_9_years` ($r = 0.939$) — the same malnutrition signal at two adjacent age bands.

One column from each pair was dropped (`infant_deaths`, `thinness_5_9_years`).

4.2 Feature Selection

We combined three independent scoring methods — **Pearson correlation** (linear relationships), **mutual information** (non-linear relationships), and **Random Forest feature importance** (interaction effects) — and dropped only features that ranked in the bottom quartile on **all three** methods simultaneously, to avoid any single method's blind spots driving the decision. Only `measles` met that bar and was dropped. `hiv_aids`, `income_composition`, `adult_mortality`, and `schooling` were consistently the top-ranked features across all three methods (Figure 6).

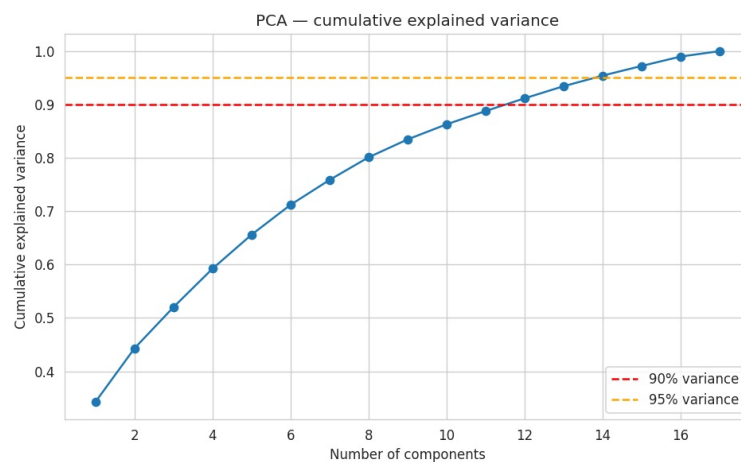


Random Forest feature importance

country (193 categories) was excluded from modeling entirely — one-hot encoding it would add 190+ sparse columns, and target-encoding it risks leaking country-level average life expectancy directly into the model. It was retained only as a non-feature identifier column, used later in error analysis.

4.3 Dimensionality Reduction (PCA)

PCA was run on the standardized, feature-selected set. **12 of the 17 components were needed to reach 90% cumulative explained variance** (14 for 95%) (Figure 7) — meaningful compression is limited because the redundant pairs were already removed in Section 4.1.



PCA cumulative explained variance

Decision: PCA components were not used in the final model.

With only a 17→12 compression at 90% variance, PCA's main benefit here is small, while its cost is real: PCA components are linear combinations of standardized features and are not interpretable, which directly conflicts with Research Question 1 (identifying *which* factors matter). PCA is documented here to satisfy the dimensionality-reduction requirement, but the modeling pipeline uses the 17 selected original features.

Final feature set (17 features): year, adult_mortality, alcohol, percentage_expenditure, hepatitis_b, bmi, under_five_deaths, polio, total_expenditure, diphtheria, hiv_aids, gdp, population, thinness_10_19_years, income_composition, schooling, status_developed.

5. Modeling and Validation

Full detail: notebooks/04_modeling_and_tuning.ipynb.

5.1 Split Strategy

Data was split **80% train / 20% held-out test** (`random_state=42`). Hyperparameter tuning used **5-fold (Ridge) or 3-fold (Random Forest, Gradient Boosting) cross-validation within the training set** as the validation layer — the test set was evaluated exactly once, at the end, and never used during model selection or tuning. All feature scaling was fit inside `sklearn.Pipeline` objects so it is refit correctly on each fold, preventing data leakage.

5.2 Model Families

Three models spanning two distinct families were trained and tuned:

- **Ridge Regression** (linear family, L2-regularized)
- **Random Forest** (bagging ensemble of decision trees)
- **Gradient Boosting** (boosting ensemble of decision trees)

5.3 Hyperparameter Tuning

Model	Method	Search space	Best parameters
Ridge	Grid search, 5-fold CV	$\alpha \in \{0.01, 0.1, 1, 5, 10, 50, 100\}$	$\alpha=50$
Random Forest	Randomized search (12 iters), 3-fold CV	$n_estimators \in \{100, 200, 300\}$, $max_depth \in \{None, 8, 15\}$, $min_samples_split \in \{2, 5, 10\}$, $min_samples_leaf \in \{1, 2, 4\}$, $max_features \in \{sqrt, log2, None\}$	$n_estimators=200$, $max_depth=15$, $min_samples_split=10$, $min_samples_leaf=2$, $max_features=None$
Gradient Boosting	Randomized search (12 iters), 3-fold CV	$n_estimators \in \{100, 150, 200\}$, $learning_rate \in \{0.01, 0.05, 0.1, 0.2\}$, $max_depth \in \{2, 3, 4\}$, $subsample \in \{0.7, 0.85, 1.0\}$	$n_estimators=200$, $learning_rate=0.1$, $max_depth=4$, $subsample=0.85$

Randomized (rather than exhaustive grid) search was used for the two ensemble models because their combined search spaces are large (up to 405 combinations for Random Forest); random search covers the space broadly at a fraction of the computational cost, which matters in a single-core environment.

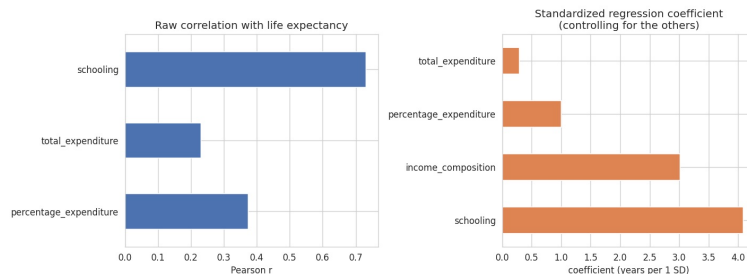
6. Answering the Three Research Questions

Full detail and code: `notebooks/06_research_questions.ipynb`. This section answers the three questions directly from the cleaned data, independent of any one model's assumptions.

6.1 Q1 — Healthcare Expenditure (% GDP) vs. Schooling

The dataset does not contain a column literally named “healthcare expenditure as % of GDP.” The two available proxies are `percentage_expenditure` (health expenditure as a % of GDP *per capita*, per the source’s own definition) and `total_expenditure` (government health spending as a % of *total government* spending, not GDP). We report both rather than overclaiming precision the data doesn’t have.

Variable	Raw correlation with life expectancy	Standardized regression coefficient (controlling for the others + income)
schooling	0.732	4.09
income_composition (control)	—	3.01
percentage_expenditure	0.374	1.00
total_expenditure	0.231	0.29

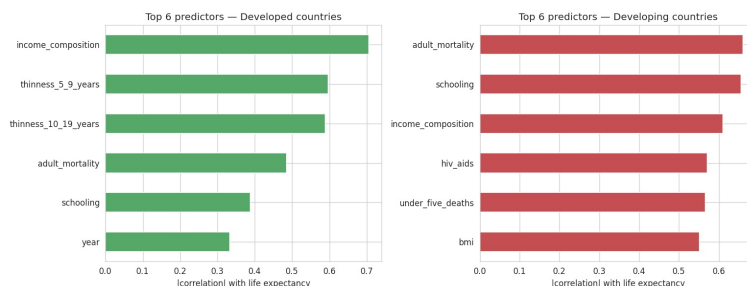


Healthcare expenditure vs. schooling

Answer: Schooling correlates with life expectancy roughly 2–3× more strongly than either healthcare expenditure measure, and remains the stronger predictor even after controlling for income — meaning the relationship isn’t just both variables tracking overall wealth. Healthcare expenditure’s apparent effect shrinks further once income is controlled for, consistent with spending partly acting as a stand-in for a country simply being wealthier, whereas schooling’s association holds up independently.

6.2 Q2 — Do Developed vs. Developing Countries Show Different Key Predictors?

We split the data by `status_developed` and computed both correlation and separate Random Forest feature importances for each group (512 Developed rows, 2,416 Developing rows).



Top predictors, Developed vs. Developing

	Developed (top predictors)	Developing (top predictors)
By correlation	income_composition (0.70), thinness measures (~0.59), adult_mortality (0.49)	adult_mortality (0.66), schooling (0.66), income_composition (0.61)
By Random Forest importance	adult_mortality (0.48), income_composition (0.23)	hiv_aids (0.60) , adult_mortality (0.18)

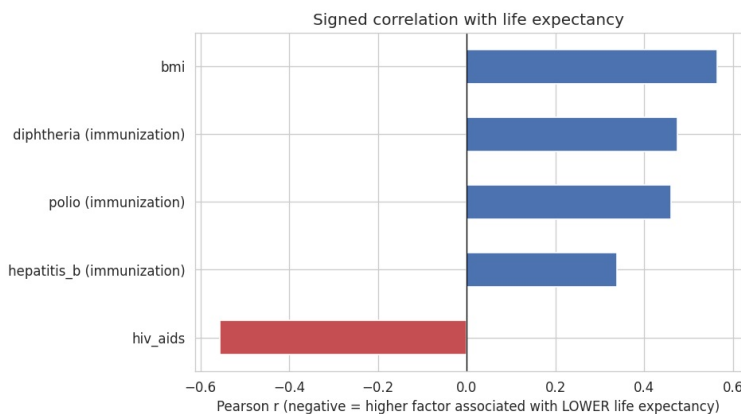
Answer: Yes, they differ, and the clearest signal comes from HIV/AIDS. By Random Forest importance, `hiv_aids` alone accounts for 60% of predictive importance for Developing countries — by far the single most important predictor — while it is essentially irrelevant for Developed countries (near-zero correlation, absent from their top predictors entirely; HIV/AIDS prevalence is almost uniformly at the

dataset's minimum in that group). Disease burden is a live, differentiating factor for Developing countries in a way it simply isn't for Developed ones. *Caveat: the Developed subgroup has far fewer rows, so its estimates are noisier and should be read as suggestive.*

6.3 Q3 — Which Single Factor Has the Largest Negative Effect?

Comparing signed correlation with life expectancy across the three candidate factor types:

Factor	Correlation with life expectancy
hiv_aids	-0.557
hepatitis_b (immunization)	+0.339
polio (immunization)	+0.460
diphtheria (immunization)	+0.475
bmi	+0.564



Signed correlation of candidate negative factors

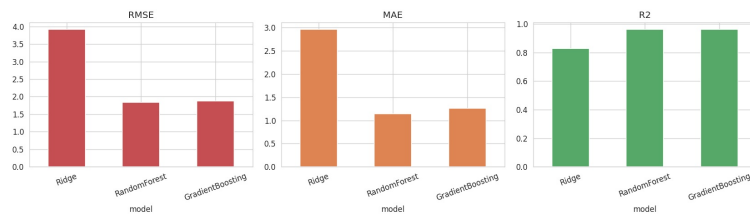
Answer: HIV/AIDS prevalence, by a wide margin — it is the only factor among the three candidate types with a negative relationship to life expectancy at all. Immunization coverage (hepatitis B, polio, diphtheria) and BMI are all positively correlated with life expectancy in this data, so they don't qualify as "negative effect" factors here. One honest caveat: BMI's positive correlation is somewhat counterintuitive given real-world evidence that very high BMI is a health risk; this dataset's BMI column has known data-quality issues in the source, so we would not treat BMI's sign here as a reliable real-world finding without further scrutiny of that specific column.

7. Results, Visualization, and Error Analysis

Full detail: notebooks/05_evaluation_and_error_analysis.ipynb.

7.1 Test Set Performance

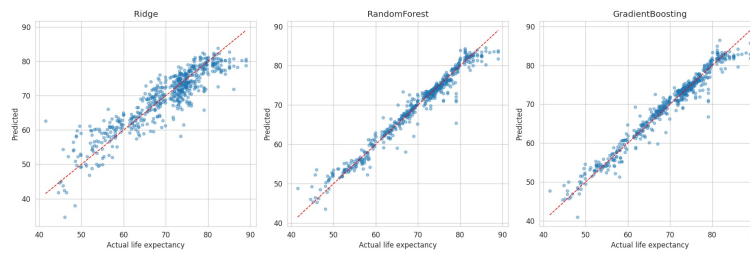
Model	RMSE (years)	MAE (years)	R ²
Ridge	3.92	2.97	0.827
Random Forest	1.84	1.15	0.962
Gradient Boosting	1.87	1.26	0.961



Metric comparison across models

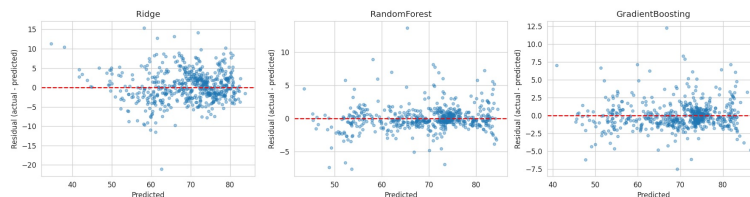
Random Forest is the best model, narrowly ahead of Gradient Boosting and substantially ahead of Ridge. The large gap between Ridge and the two tree ensembles (R^2 0.827 vs ~ 0.96) directly answers part of Research Question 3: linear regression, however regularized, cannot capture the true relationship between these indicators and life expectancy — the relationship is non-linear.

7.2 Actual vs. Predicted and Residuals



Actual vs. predicted, all three models

Ridge visibly deviates from the diagonal below ~ 55 years of life expectancy, systematically **overpredicting** life expectancy for the worst-off countries — it cannot capture how severe outcomes get in crisis-level cases. Both tree ensembles track the diagonal closely across the full range.



Residual plots, all three models

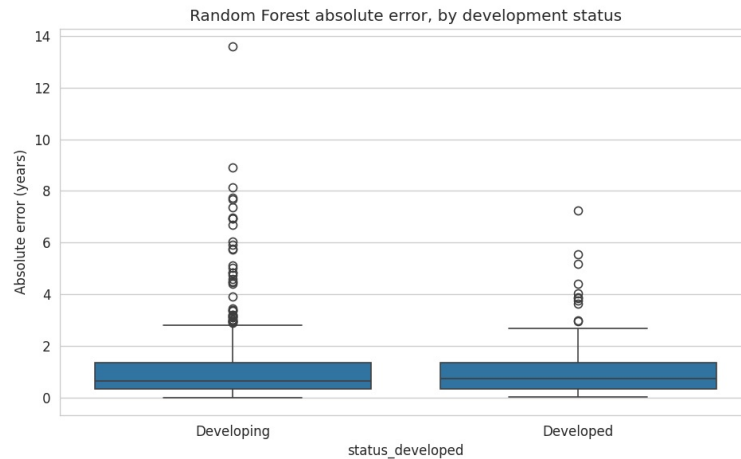
Ridge's residuals show a curved, funnel-like pattern rather than random scatter around zero — a textbook sign of a misspecified (too-simple) model. Random Forest and Gradient Boosting residuals are much closer to a random band around zero, with slightly more spread at the low end.

7.3 Where the Best Model Fails

Examining the 15 largest errors made by Random Forest on the test set revealed two distinct causes, rather than one:

1. **Extreme HIV/AIDS-prevalence countries are underestimated in severity.** Swaziland ($\sim 50\%$ HIV/AIDS prevalence, actual life expectancy 45.6–45.9 years) and Sierra Leone (2003, actual 41.5 years) were all predicted 6–8 years *higher* than reality. The model correctly learns that high HIV/AIDS prevalence lowers life expectancy, but underestimates the magnitude of the effect at the most extreme end — likely because very few country-years in the training data sit at that extreme.
2. **A data-quality artifact inherited from the source, not a true model failure.** Saint Vincent and the Grenadines reports the identical value, 79.0 years, for three consecutive years (2000–2002) — a strong sign the original data carried forward a stale estimate rather than a fresh yearly measurement. The model's predictions (65.4–71.3) are informed by that country's actual indicator values, which understandably don't match a figure that wasn't really re-measured each year.

We also tested our initial hypothesis that Developing countries would show higher error than Developed countries (Figure 8) — **this was not supported by the data**: mean absolute error was nearly identical (1.14 years for Developing vs. 1.18 years for Developed). What actually predicts a large error is being an extreme outlier on a specific indicator (very high HIV/AIDS prevalence), not a country's broad development category. This directly informs Research Question 2: the Developed/Developing gap in this dataset is *not* primarily driven by unmodeled category-level factors — the measurable indicators already explain most of it, aside from a small number of extreme, indicator-specific cases.



Random Forest absolute error by development status

8. Conclusion

Beyond the three questions above, we also built and evaluated a predictive model as a secondary exercise: Random Forest, tuned via randomized search with cross-validation, achieved the best test-set performance (RMSE 1.84 years, R^2 0.962), well ahead of a tuned linear model (Ridge, R^2 0.827) — showing the relationships between these indicators and life expectancy are meaningfully non-linear.

Answering our three research questions (Section 6):

- Schooling correlates with life expectancy roughly 2-3× more strongly than healthcare expenditure** ($r \approx 0.73$ vs. $r \approx 0.37/0.23$ for the two expenditure proxies available), and remains the stronger predictor even after controlling for income.
- Yes, Developed and Developing countries show different key predictors.** The clearest signal: HIV/AIDS prevalence is the single dominant predictor for Developing countries (Random Forest importance 0.60) but is essentially irrelevant for Developed countries.
- HIV/AIDS prevalence has by far the largest negative effect** among the three candidate factor types ($r \approx -0.56$) — it's the only one of the three with a negative relationship to life expectancy at all; immunization rates and BMI are both positively correlated in this data.

Limitations: the dataset only runs through 2015, has no measure of political stability or conflict intensity, and — as shown above — contains at least some values that are not fresh yearly measurements. Any deployment of this model for real-world decision-making should treat predictions for extreme, low-life-expectancy cases with wider uncertainty than the aggregate R^2 suggests.

Appendix A: Repository Structure and Reproduction

cse437-life-expectancy-group1/
└─ README.md

```

├── requirements.txt
├── .gitignore
├── data/
│   ├── raw/                # original, untouched CSV
│   └── processed/          # cleaned + feature-engineered
output
├── README.md
├── notebooks/
│   ├── 01_data_audit_and_eda.ipynb
│   ├── 02_preprocessing.ipynb
│   ├── 03_feature_engineering.ipynb
│   ├── 04_modeling_and_tuning.ipynb
│   ├── 05_evaluation_and_error_analysis.ipynb
│   └── 06_research_questions.ipynb
├── src/
│   └── utils.py             # shared loading / metrics
functions
├── models/                 # saved .joblib model files +
results CSV
├── figures/                # every figure referenced in this
report
├── report/
│   ├── report.md
│   └── report.pdf

```

To reproduce all results:

1. Clone the repository and create a virtual environment.
2. `pip install -r requirements.txt`
3. Run the five notebooks in notebooks/ **in numeric order**, top to bottom, on a fresh kernel. Each notebook reads its input from data/ or a prior notebook's saved output, and writes its own output back to data/processed/, models/, or figures/ — so the numeric order matters.
4. All relative paths assume the notebook is run from inside the notebooks/ folder (VS Code's Jupyter extension does this automatically when you open a .ipynb file directly).

Appendix B: Full List of Original Features

country, year, status, life expectancy, adult mortality, infant deaths, alcohol, percentage expenditure, hepatitis B, measles, BMI, under-five deaths, polio, total expenditure, diphtheria, HIV/AIDS, GDP, population, thinness 1-19 years, thinness 5-9 years, income composition of resources, schooling — 22 columns total in the raw file, reduced to 17 predictive features plus the target after the cleaning and feature-selection process described in Sections 2 and 4.